

PFB Scalping, Noise Bandpass, and Narrowband Signal Intensity

setigen voltage response investigation
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Abstract. The coarse-channel scalping visible in voltage-generated spectrograms is not a single universal correction factor. A deterministic narrowband tone and white Gaussian voltage noise pass through the same polyphase filterbank (PFB), but the detected quantities are different. A tone centered on a final fine-channel bin is attenuated by the squared response of one PFB analysis branch. The fine-channelized noise baseline follows an aliased power sum of neighboring PFB branches. These two transfer functions agree well in the middle of a coarse channel, but they differ strongly at the coarse-channel edge. In the default experiment, the modeled noise bandpass at the edge is about 0.518 of the coarse-channel center, while the single-branch tone transfer is about 0.250. The voltage simulation matches the single-branch prediction to a maximum absolute error of 1.7×10^{-5} over the tested fine-bin positions.

1 Question

We want to know how a fixed original narrowband signal intensity maps into a detected spectrogram intensity as the signal moves across the fine-channel bins inside one coarse channel. The practical calibration question is:

$$I_{\text{detected}}(f) - I_{\text{modeled bandpass}}(f) \quad \text{versus} \quad I_{\text{original}}.$$

This matters for synthetic voltage injection and for interpreting real observations, because the instrumental PFB response can change the apparent signal excess independently of natural sky continuum structure.

2 Two Different Transfer Functions

Let B be the number of PFB branches, T the number of taps, and $h[n]$ the prototype PFB window/filter coefficients for $0 \leq n < TB$. Consider a tone whose frequency is offset by δ coarse-channel widths from the center of analysis branch k :

$$x[n] = Ae^{2\pi i(k+\delta)n/B}.$$

Ignoring phase and normalization constants that cancel in ratios, the complex amplitude response in branch k is the prototype transform

$$G(\delta) = \sum_{n=0}^{TB-1} h[n]e^{2\pi i\delta n/B}.$$

If the downstream fine FFT is centered exactly on the tone, there is no fine-FFT sinc leakage. The detected tone power in that final channel is therefore

$$R_{\text{tone}}(\delta) = \frac{|G(\delta)|^2}{|G(0)|^2}.$$

White voltage noise is different. After critical decimation by the PFB, the power spectral density of one channelized branch includes aliased contributions from frequencies separated by coarse-channel widths:

$$R_{\text{noise}}(\delta) \propto \sum_m |G(\delta + m)|^2,$$

For this PFB, the adjacent terms dominate near a coarse-channel edge. This is why the observed noise baseline, and the existing `PolyphaseFilterbank.get_response()` / `tile_response()` curve, follow a smoother aliased response. A deterministic tone in the target final channel does not receive the same aliased power sum. It follows the single-branch response.

3 What “Single Analysis Branch” Means

A PFB analysis channel is one row of a uniform filterbank. Operationally, the input voltage stream is multiplied by a prototype filter, folded into B phases, summed, and then Fourier transformed across those phases. The output branch k is therefore not an infinitely sharp rectangular spectral bin. It is one bandpass filter centered on coarse-channel k , with a response set by the prototype coefficients and modulation phase.

For a deterministic tone, the voltage entering branch k adds coherently. If the tone is at frequency $k + \delta$ in coarse-channel units, every sample has a fixed phase relationship. After the PFB sum, the target-branch complex amplitude is $A G(\delta)$ up to a phase and normalization. Power detection then gives

$$P_{\text{tone},k}(\delta) = C_{\text{sig}} |A|^2 |G(\delta)|^2,$$

where C_{sig} absorbs the FFT normalization, integration length, polarization convention, and any quantization scale. This is the single-branch response: one physical analysis filter acting on one coherent tone. If the fine FFT bin is exactly centered on that tone, no additional sinc leakage appears in the fine FFT.

The noise baseline is a different statistical object. For white voltage noise, the input has independent random phases across all frequencies. Critical decimation by the PFB aliases spectral images separated by one coarse-channel spacing into the same branch. These random components do not add coherently in voltage, but their variances add in power:

$$E[P_{\text{noise},k}(\delta)] = C_{\text{noise}} S_0 \sum_m |G(\delta + m)|^2,$$

where S_0 is the white-noise spectral density. Thus the noise scalloping is a variance response, while the narrowband signal excess is a coherent single-filter response. The same PFB window creates both curves, but the measured quantities are not the same.

This also explains why the edge behavior is asymmetric with intuition from a smooth bandpass. At the exact boundary between two coarse channels, two neighboring analysis filters each see part of the broadband noise variance. The noise baseline in a final channel is therefore not as low as the target-branch response to a tone. A tone assigned to the target branch near that same boundary is attenuated by the target branch alone, so its detected peak excess falls more steeply.

4 Predicting Detected Power as a Function of Frequency

For synthesis and calibration, the most useful formulation is a detector equation. Let a signal have intrinsic voltage amplitude $A(t)$ and instantaneous frequency $f(t)$. Let $f_{c,k}$ be the center of the nearest or selected coarse analysis branch, and define

$$\delta(t) = \frac{f(t) - f_{c,k}}{\Delta f_{\text{coarse}}}.$$

For an exact-bin, zero-drift tone measured with a peak-channel detector, the expected detected excess above the modeled bandpass is

$$E[\Delta P_{\text{peak}}] = C_{\text{sig}} |A|^2 R_{\text{tone}}(\delta),$$

where

$$R_{\text{tone}}(\delta) = \frac{|G(\delta)|^2}{|G(0)|^2}.$$

In practice C_{sig} can be fixed by a center-channel calibration tone:

$$C_{\text{sig}} = \frac{E[\Delta P_{\text{center}}]}{|A_{\text{center}}|^2}.$$

Then the original voltage-amplitude scale follows

$$|A|^2 \approx \frac{\Delta P_{\text{peak}}}{C_{\text{sig}} R_{\text{tone}}(\delta)}.$$

This is a power-domain correction. A voltage-amplitude correction uses the square root of the power correction.

For off-bin tones, the fine FFT adds its own response. If the tone is offset by ϵ fine-bin widths from the final FFT bin center, a rectangular fine FFT gives the familiar Dirichlet/sinc factor:

$$R_{\text{fine}}(\epsilon) \approx \left| \frac{\sin(\pi\epsilon)}{N \sin(\pi\epsilon/N)} \right|^2 \approx \text{sinc}^2(\epsilon)$$

for large FFT length N . A first-order peak-channel model is then

$$E[\Delta P_{\text{peak}}] \approx C_{\text{sig}} |A|^2 R_{\text{tone}}(\delta) R_{\text{fine}}(\epsilon).$$

If the detector sums a local final-frequency aperture $\mathcal{A}$ rather than one peak channel, replace the peak response by an aperture response:

$$R_{\mathcal{A}}(\delta, \epsilon) = \sum_{j \in \mathcal{A}} R_{\text{tone}}(\delta_j) R_{\text{fine}}(\epsilon_j),$$

with the same normalization convention as the measurement. For drifting tones, the final prediction is a time average along the path:

$$E[\Delta P_{\mathcal{A}}] \approx \frac{C_{\text{sig}}}{N_t} \sum_{i=1}^{N_t} |A(t_i)|^2 R_{\mathcal{A}}(\delta(t_i), \epsilon(t_i)),$$

plus any explicit dedrift/integration weights used by the search pipeline.

The noise model is separate:

$$E[P_{\text{baseline}}(f)] = C_{\text{noise}} S(f) R_{\text{noise}}(\delta), \quad R_{\text{noise}}(\delta) = \frac{\sum_m |G(\delta + m)|^2}{\sum_m |G(m)|^2}.$$

Here $S(f)$ is the smooth continuum/sky/instrument temperature term. In real data, $S(f)$ can vary across the band because of receiver gain, sky continuum, and RFI occupancy. The PFB scalloping term is the repeated instrumental factor inside that broader continuum.

5 How to Use This for Accurate Prediction

For synthetic voltage work, the prediction can be exact for the implemented pipeline:

1. Read the actual PFB prototype coefficients $h[n]$.
2. Compute $G(\delta)$ on a dense grid across one coarse channel.
3. Compute $R_{\text{tone}}(\delta)$ for the signal detector aperture.
4. Compute $R_{\text{noise}}(\delta)$ separately for the baseline model.
5. Calibrate the overall scale with a center-bin tone, or derive it from the pipeline normalization if every normalization factor is tracked.

For setigen, this means a response-aware voltage synthesis or analysis helper should not reuse `get_response()` as the narrowband correction. It should expose both a noise response and a tone response.

For real observations, the same mathematical structure applies, but the instrument must be matched:

- The PFB tap count, prototype window, and normalization must match the instrument or reduction pipeline.
- Any requantization, clipping, gain normalization, or bandpass flattening can change the observed scale.
- The measured continuum can constrain the repeated noise response, but it does not uniquely determine the tone response unless the filter family is known or assumed.
- A strong unresolved narrowband calibrator, injected tone, or synthetic voltage round-trip is the cleanest way to validate the signal-transfer response.

Thus we can accurately predict power as a function of coarse-channel position when we know the analysis filterbank and the detector aperture. If the real instrument response is only approximately known, the prediction should be reported with model uncertainty and validated against observed or injected tones.

6 Experiment

The reproducible experiment is implemented in `response_experiments.py`. The default configuration is:

$$T = 8, \quad B = 64, \quad F = 256, \quad \Delta f_{\text{coarse}} = 1 \text{ MHz}, \quad \Delta f_{\text{fine}} = 3906.25 \text{ Hz}.$$

The script computes R_{tone} directly from the PFB window, compares it against clean voltage simulations of exact-bin zero-drift tones, and then runs noisy simulations in which a scaled ideal PFB bandpass is fit and subtracted.

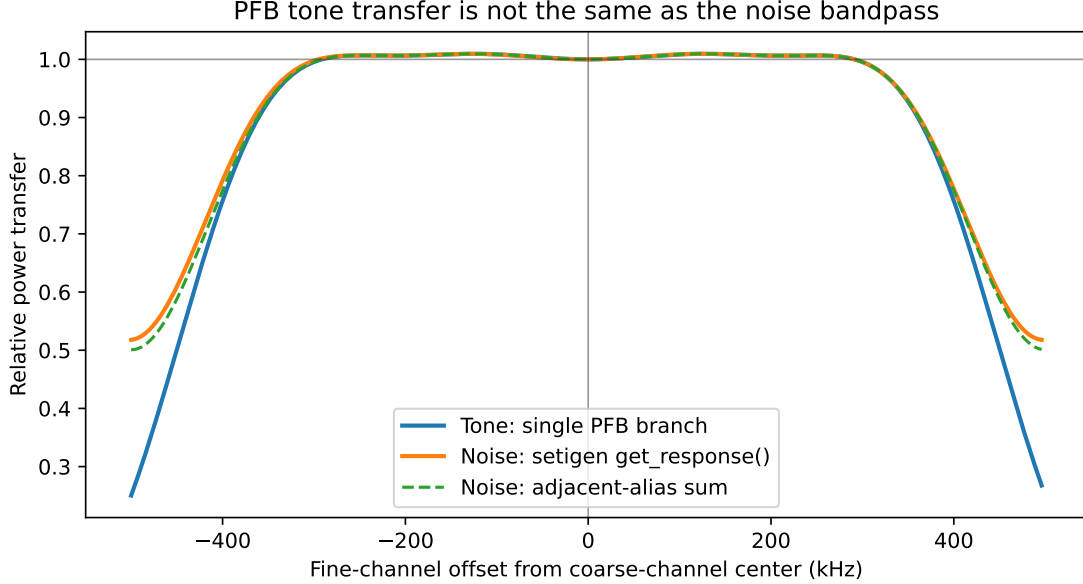


Figure 1: The single-branch tone transfer and the aliased noise bandpass are different transfer functions. At the coarse-channel edge, the noise bandpass remains near 0.52 of center, while the deterministic tone branch response is near 0.25 of center.

7 Numerical Summary

For the exact coarse-channel edge in the default experiment:

single-branch tone prediction	0.2504,
modeled noise bandpass	0.5179,
noisy peak excess above modeled bandpass	0.2433 ± 0.0117 ,
noisy ± 3 final-channel aperture excess	0.2140 ± 0.0336 .

All quantities are normalized to a tone at the center of the same coarse channel. The peak-channel detected excess agrees with the single-branch PFB prediction, not with the noise-bandpass response.

8 Interpretation

The coarse-channel edge discrepancy is expected signal-processing behavior, not a numerical accident. The PFB is critically sampled. Noise is stochastic and broadband, so after decimation its fine-channelized baseline contains aliased power from adjacent spectral images of the prototype filter. A narrowband tone at a known frequency is not a broadband stochastic field. In the target analysis branch, its detected power is controlled by the one branch response at that frequency. Therefore a correction curve inferred from the noise bandpass is not sufficient to recover the original signal intensity of a narrowband tone.

Equivalently:

$$I_{\text{noise baseline}}(f) \propto R_{\text{noise}}(f), \quad I_{\text{tone excess}}(f) \propto R_{\text{tone}}(f).$$

Near the center of a coarse channel these responses are similar enough that the difference is small. Near a coarse-channel edge they diverge by roughly a factor of two in power for this configuration.

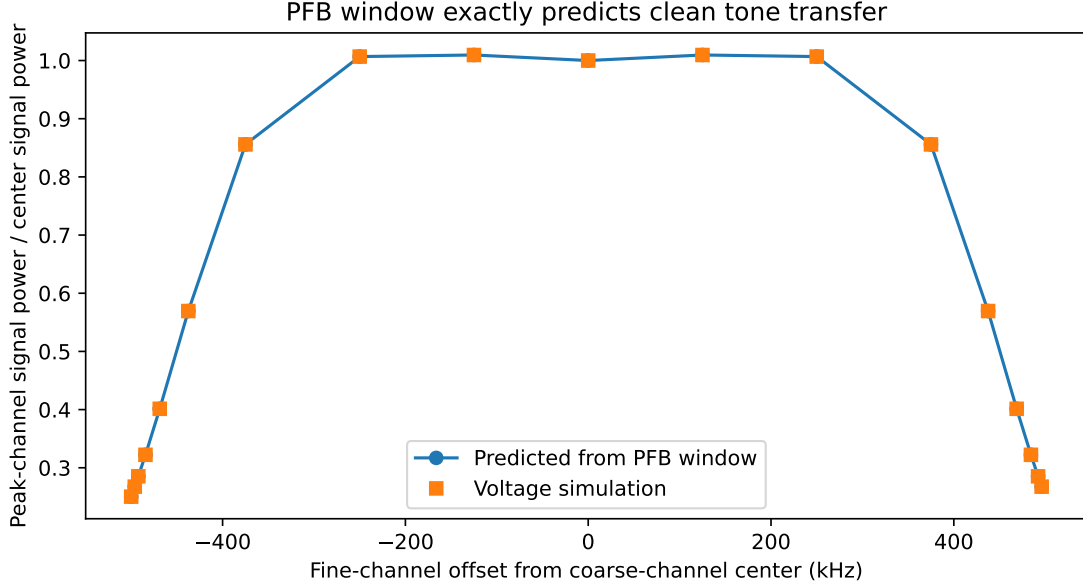


Figure 2: The PFB-window prediction for clean tone transfer matches the voltage simulation. The maximum absolute error over the tested bins is 1.7×10^{-5} in center-normalized power.

9 Implications for setigen and Real Observations

The voltage pipeline is accurately modeling the PFB that it implements: the same PFB window predicts the measured tone response essentially exactly. The remaining question is whether the implemented PFB parameters and downstream reduction path match a given real instrument and product. For real observations, the calibration depends on the actual PFB coefficients, tap count, coarse-channel selection, quantization/requantization behavior, fine FFT, normalization, and any later bandpass flattening.

For scientific use, we should maintain two separate concepts:

1. A noise-bandpass response, useful for modeling local background and PFB scalloping in noise statistics.
2. A narrowband signal-transfer response, useful for recovering original signal intensity from detected excess.

Using the noise-bandpass model alone would overestimate the recovered original intensity of edge-channel narrowband signals. The appropriate correction for a peak-channel detector is approximately

$$I_{\text{original}} \approx \frac{I_{\text{detected}} - I_{\text{modeled bandpass}}}{R_{\text{tone}}(\delta)},$$

with additional factors for off-bin fine-FFT leakage, drift smearing, and the chosen detection aperture.

10 Recommended Next Steps

- Expose or document both response curves in the voltage tooling: `noise_response` and `tone_response`.
- Repeat this experiment through the full RAW write/read/reduce path and with quantization enabled.
- Compare the implemented PFB window and response against rawspec and known instrument configurations.
- Add off-bin and drifting tones so the correction includes fine-FFT sinc leakage and drift-smearing effects.
- For real observations, estimate continuum/bandpass shape locally but correct narrowband signal intensity using the signal-transfer curve, not the noise-bandpass curve.

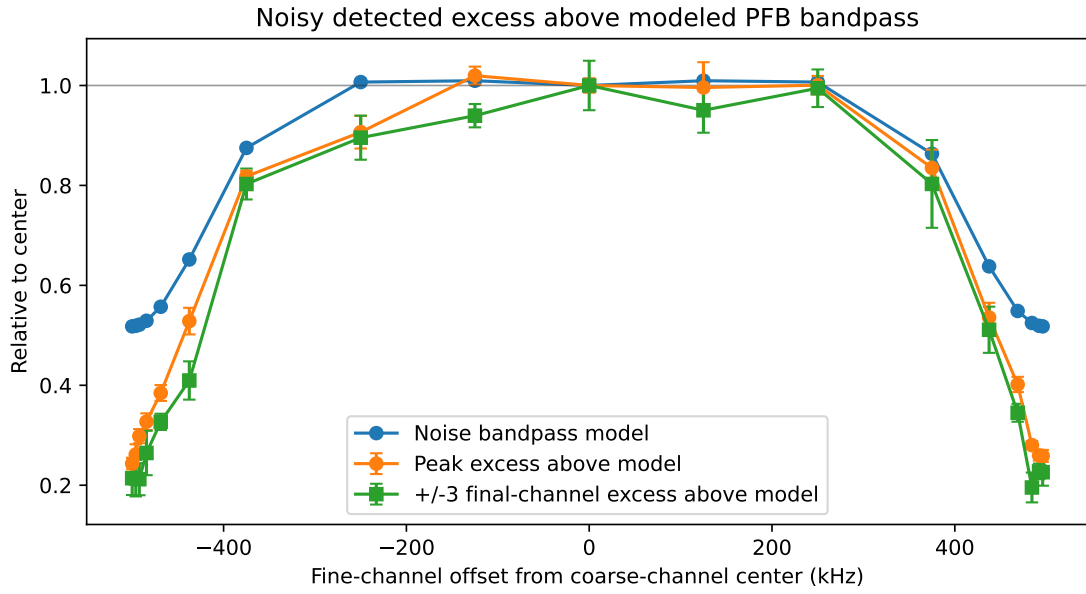


Figure 3: Noisy detected excess after subtracting a scaled modeled PFB bandpass. The edge tone is weaker than one would infer from the noise bandpass alone, matching the single-branch signal-transfer prediction.